Siniša Košćec

Senior Full Stack Software Engineer | AI Engineering

Varaždin, Croatia · +385 97 698 4484
sinisa@dvasadva.com | LinkedIn | dvasadva.com | GitHub

SUMMARY

Senior full stack engineer with almost 20 years of experience, combining production web and mobile engineering with AI and LLM engineering. Started at the intersection of finance and IT in SAP ERP systems, then spent 10+ years across the full web and mobile pipeline, from architecture and prototyping to frontend and backend API engineering with React, Next.js, React Native, Node.js and TypeScript. The last seven years have centered on machine learning, NLP and LLM engineering, shipping production AI systems rather than experiments: RAG platforms, tool-using AI agents, guardrailed chatbots on Amazon Bedrock, and multi-model applications.

Tool and vendor agnostic, with hands-on work across Anthropic Claude, OpenAI, Google Gemini and open models (Hugging Face, Together AI), using direct APIs, the Vercel AI SDK, LangChain and cloud platforms such as AWS Bedrock. Solid grasp of LLM application patterns: retrieval-augmented generation, agents and tool use via MCP, evaluation and tracing (Langfuse, Weights & Biases) and guardrails. Strong interest in self-hosted and open-weight models: runs quantized DeepSeek, Qwen and other open models locally with llama.cpp and vLLM, weighing hardware, quantization, latency, privacy and cost against hosted APIs.

Fully AI-native in daily engineering. Works with Claude Code, Codex, Cursor, Devin, Windsurf and GitHub Copilot every day, and builds and wires up MCP servers, custom agent skills, slash commands, subagents and hooks. Runs multiple coding agents in parallel on isolated Git worktrees for implementation, code review, test generation and refactoring, and closely follows the newest agent tooling, including agent harnesses such as Pi, durable long-running agents (Pi Durable) and stateful agents on Cloudflare Durable Objects.

Enjoys researching and comparing tools, finding where AI actually pays off, sketching architectures in Excalidraw, and explaining designs and tradeoffs clearly to both developers and leadership.

TECHNICAL STACK

Frontend & Mobile	JavaScript / TypeScript, React.js, Next.js, React Native (Expo), Astro, TanStack Start, TanStack Query, Zustand, HTML5, CSS3, Tailwind CSS, NativeWind, Shadcn/ui
Backend & APIs	Node.js, Nest.js, Express.js, Fastify.js, Python (FastAPI, Flask), Ruby on Rails, GraphQL, REST API design, WebSockets
Architecture	Software, application and cloud architecture, solution design, event-driven systems (Apache Kafka, RabbitMQ)
Database & Sync	PostgreSQL (Neon), MySQL, MongoDB, Redis (Upstash), Supabase, Cloud Firestore, TanStack DB, ElectricSQL
Cloud & DevOps	AWS (S3, CDK, Bedrock), Docker, Nginx, Linux, Bash, GitHub Actions, Vercel, Railway, Firebase, Convex, Cloudflare R2
Testing & Quality	Jest, Vitest, Playwright, Selenium, Sentry, LogRocket

AI / LLM	RAG, AI agents, MCP, LangChain, LangGraph, LlamaIndex, Vercel AI SDK, Anthropic API, OpenAI API, Azure OpenAI, Google Gemini, Hugging Face, Together AI
LLMOps & Vectors	ChromaDB, Qdrant, Weaviate, Supabase pgvector, Langfuse, Weights & Biases
ML & Inference	PyTorch, PyTorch Lightning, scikit-learn, computer vision, CUDA, llama.cpp, vLLM, quantized open models (DeepSeek, Qwen)
Agentic Dev Tools	Claude Code, Codex, Cursor, Devin, Windsurf, GitHub Copilot, MCP servers, custom skills, subagents, hooks, parallel agents on Git worktrees
Tools	Git, GitHub, Figma, Excalidraw, Scrapy, Beautiful Soup
Languages	Croatian, English

EXPERIENCE

Dvasadva d.o.o. **August 2016 – Present**
Founder & CEO *Varaždin, Croatia*

Specializing in Full Stack development and LLM engineering. Delivering end-to-end solutions ranging from mobile applications to complex AI-driven web platforms.

Key Recent Projects:

1. Appointment Booking Platform – Mobile App & Supporting Services *Dec 2025 – present*
Part of an engineering team building a cross-platform appointment booking app on a distributed, event-driven architecture; owning the Expo mobile app and supporting web apps while collaborating with the backend team.

- **Tech Stack:** React Native (Expo), TypeScript, Ruby on Rails, GraphQL, PostgreSQL (Neon), self-hosted Apache Kafka, TanStack Start, TanStack DB, ElectricSQL.
- **Architecture:** Mobile client communicates with a Ruby on Rails backend over GraphQL; the team operates an event-driven backbone using self-hosted Kafka (Docker) with multiple dedicated consumers for billing, logging, and tracing.
- **Supporting Web Apps:** Contributed to side applications that assist the main app, using TanStack Start and TanStack DB, with a Neon PostgreSQL database and an ElectricSQL sync layer for local-first, real-time data synchronization.
- **Observability:** Worked with Sentry and LogRocket for error tracking, session replay, and end-to-end tracing across the stack.
- **Workflow:** Developed in a fully AI-native, collaborative environment, using agentic tools (Claude Code, Codex), MCP servers, and custom skills daily across implementation, debugging, and refactoring.

2. RAG Platform – Documentation Intelligence (Infodesign d.o.o.) *Feb 2024 – Mar 2025*
A Retrieval-Augmented Generation platform for internal documentation search across a Croatian ERP software company.

- **Context:** Infodesign develops ERP solutions for municipal bodies and utility companies. Documentation was decentralized, often obsolete, and partially embedded in e-mail threads between key users and engineers.
- **Tech Stack:** Python, LangChain, ChromaDB, FastAPI, OpenAI API, Langfuse, Weights & Biases.
- **Ingestion Pipeline:** Built multi-source loaders for `.docx` files (Docx2txt), PDFs (PyPDF), CSV tables,

and e-mail archives; generated embeddings using OpenAI embedding models.

- **Retrieval Optimization:** Experimented with chunk sizes, multiquery retrieval, and ensemble retrievers to maximize accuracy on tabular and prose documentation.
- **Observability:** Integrated Langfuse and Weights & Biases for end-to-end pipeline tracing and evaluation.
- **Model Strategy:** Progressed from OpenAI GPT models toward open-source alternatives (HuggingFace and Together AI) served via commercial APIs, addressing the client's privacy and cost requirements.

3. Juicefast AI – Wellness & Nutrition Mobile App *Jan 2025 – Dec 2025*
A cross-platform React Native application. (App Store | Google Play)

- **Tech Stack:** React Native (Expo), TypeScript, Expo Router, NativeWind, Convex, Clerk.
- **AI Integration:** Integrated Vercel AI SDK for meal recognition, analyzing food photos to return structured JSON objects with accurate macronutrients.
- **Billing & Subscriptions:** Integrated RevenueCat for In-App Purchases (IAP) and built a custom API to synchronize subscription statuses between web and mobile.
- **UX/UI:** Implemented optimistic UI patterns (`useOptimistic` hook) for zero-latency wellness tracking.
- **Real-time:** Built an AI chat assistant and a live messaging system connecting users with nutritionists.

4. edureteAI – Educational Chatbot Arena *Apr 2024 – Dec 2025*
An educational platform enabling students to compare LLM outputs. (ai.edurete.com)

- **Tech Stack:** Next.js (App Router), Node.js, Vercel AI SDK, TypeScript.
- **AI Orchestration:** Integrated multimodal models (OpenAI, Anthropic, Google) for comparative analysis.
- **Data & State:** Utilized Upstash Redis for chat history and Neon Serverless PostgreSQL for billing limits.
- **Agent Tools:** Developed a custom Python API (FastAPI/Matplotlib) serving as a tool for the AI agent to generate dynamic graphs.
- **Infrastructure:** Integrated Stripe for billing and Cloudflare R2 for image hosting.

5. Coppin State University – AI Chatbot (with Mindgrub Technologies) *Sep 2024 – Dec 2024*
A conversational assistant for the university website, built on Amazon Bedrock. (Baltimore, USA)

- **Tech Stack:** Amazon Bedrock, AWS CDK, Amazon S3, Anthropic (Claude) models, WebSockets, Drupal (source CMS).
- **Data Pipeline:** Extracted website content into an Amazon S3 bucket and cleaned/processed the HTML to produce high-quality data, then indexed it into a vector database for retrieval.
- **Safety & Augmentation:** Configured Bedrock Guardrails to keep responses on-topic and implemented Action Groups to augment the LLM with additional data sources.
- **Integration:** Connected the Bedrock-backed system to the live website via WebSockets for real-time chat.

INA Grupa **April 2007 – February 2016**
SAP Finance Module Specialist *Zagreb, Croatia*

- Business process examination and optimization in the SAP ERP suite regarding the finance department.
- Resolved key user issues concerning accounts receivable, accounts payable, and general ledger.
- Engagement on I4 project (SAP integration between INA and MOL) and I5 project (SAP ERP deployment in subsidiaries).

EDUCATION

Activeloop LLC July 2023
Certificate of Completion: LangChain & Vector Databases in Production, NLP | LLM

Faculty of Economics & Business Zagreb 2001 – 2007
Master's degree